

Stanford University Institutional Biosafety Committee

Panel 2 Minutes of Meeting August 19, 2026

Present (Voting)

M. Holodniy, MD (Chair)
S. Feldman, PhD (Co-chair)
A. Bhatt, PhD
P. Yang, PhD
R. Paulmurugan, PhD
S. Oliver, PhD
R. Trujillo, PhD
S. Chen
J. Arunachalam
S. Vleck, PhD (BSO)

Also Present (Not Voting)

K. Lin, PhD
B. Donnelly
K. Nobrega
C. Inacay, left at 5:39
S. Rayate, left at 5:39

Prior to the meeting, all IBC members should review:

- Confidentiality
 - The Panel proceedings are confidential. No protocols, including proprietary information, reviewed by Biosafety or IBC members and/or presented at Panel meetings should be discussed with anyone other than Panel members.
- Conflict of Interest
 - Any person with a conflicting interest in a protocol must leave the room during discussions and voting on the protocol. Conflicting interest includes participating in or supervising the project, a financial interest, a personal or fiduciary relationship, or some other situation giving rise to a conflicting interest as defined in the Guidelines for IBC Members on Conflicting Interests. A member who leaves the room for any reason will not be counted in the quorum for any vote that takes place during their absence.
- Designated Member Review
 - Please be reminded that all IBC members have agreed in advance, in writing, to use Designated Member Review subsequent to Full Committee Review when a modification is needed to secure approval of any of the protocols being discussed and voted on today. IBC members will have the modified research protocol available to them, and any IBC member may at any time request Full Committee Review of the protocol.

The meeting was called to order at 5:10 PM by M. Holodniy, Chair. A quorum (five or more voting members) was present. The meeting was hybrid.

Agenda Items

- a. The first order of business was a reminder that the Panel proceedings are confidential, though the meeting minutes shall be made publicly available. All protocols reviewed and/or

presented, including proprietary information, should not be discussed outside convened meetings.

- b. The second order of business was a reminder that any person with a conflicting interest in a protocol must leave the room during discussions and voting on the protocol. "Conflicting interest" includes participating in or supervising the project, an outside interest, a personal or fiduciary relationship, or some other situation giving rise to a conflicting interest as defined in the Guidelines for IBC members on Conflicting Interest. A member who leaves the room for any reason will not be counted in the quorum for any vote that takes place during their absence.
- c. The third order of business was the reminder that all IBC members have agreed in advance, in writing, to use Designated Member Review (DMR) subsequent to Full Committee Review when a modification is needed to secure approval of any of the protocols being discussed and voted on today. IBC members will have the modified research protocol available to them, and any IBC member may at any time request Full Committee Review of the protocol.
- d. The fourth order of business was review and voting on the minutes of July 15, 2026, which were distributed electronically to all IBC members prior to this meeting.
 - Voting on May minutes—approval, unanimous, no dissenters

5. The fifth order of business was IBC Business/Education

None

6. Protocols

a. Clinical

PI	Protocol
1. Frank, M.	[6239] CCT5141: A Phase 1/2, open-label, multicenter study of UB-VV400 with rapamycin in relapsed or refractory B-cell malignancies
	New Protocol Summary: The purpose of this study is to learn if the study drug called UB-VV400 can help patients with large B-cell lymphoma (LBCL) and mantle cell lymphoma (MCL). The study will also look for the best dose level and see how safe UB-VV400 is for participants. This trial will evaluate the study drug, UB-VV400, when given with another drug called rapamycin. UB-VV400, an investigational, replication-incompetent, self-inactivating (SIN) lentiviral vector (LVV), composed of a surface-engineered viral envelope with a membrane-bound multidomain fusion (MDF) protein, that encodes a transgene for an anti-CD22 chimeric antigen receptor (CAR) and a small molecule-controlled synthetic cytokine receptor (rapamycin-activated

	cytokine receptor [RACR]). UB-VV400 is pseudotyped with the envelope glycoprotein of the coxal vesiculovirus (coxal). Training: Complete Applicable Section of the NIH Guidelines: Section III-C Containment Conditions: BSL2 Special Provisions: Hospital/Clinic Infection Control precautions Discussion: Committee Members discussed mechanisms and targeting effects, as well as theoretical risk of tumor targeting and antigen downregulation. Voting: A motion was made to approve the protocol and was seconded. Total 10, For 10, Opposed 0, Abstain 0
2. Do, D.	[6242] A Multi-Center, Randomized, Double-Masked, Active-Comparator-Controlled, Phase 3 Study to Evaluate the Efficacy and Safety of ixoberogene soroparvovec (Ixo-vec) in Participants with Neovascular Age-Related Macular Degeneration (AQUARIUS)
	New Protocol Summary: Ixoberogene soroparvovec is a replication-incompetent, adeno-associated virus (AAV) vector comprising the AAV.7m8 capsid, carrying a version of codon-optimized complementary DNA (cDNA) for aflibercept, an anti-vascular endothelial growth factor (VEGF) molecule, under the control of a ubiquitous promoter. As such, a single intravitreal(ly) (IVT) injection of Ixo-vec provides sustained therapeutic levels of aflibercept via a ‘biofactory approach’ to optimize functional and anatomic outcomes, while minimizing the burden of frequent anti-VEGF IVT injections. Stable levels of aflibercept delivered via Ixo-vec have the potential to reduce real-world undertreatment and fluid fluctuations associated with bolus anti-VEGF IVT injections, and thereby, improve long-term vision preservation and quality of life for patients with neovascular (wet) age-related macular degeneration. Training: Complete Applicable Section of the NIH Guidelines: Section III-C Containment Conditions: BSL1 Special Provisions: Hospital/Clinic Infection Control precautions Discussion: A Committee Member asked if the AE involving ocular herpes had received sham or agent product, what type of herpesvirus was

	involved, and how it was determined this was unrelated. The reviewer will follow up with the PI and provide the information to the IBC members. Voting: A motion was made to approve the protocol and was seconded. Total 10, For 10, Opposed 0, Abstain 0
3. Shkolyar, E.	[6283] ResQ132EX-NMIBC: Expanded Access Use of Recombinant Bacillus Calmette-Guérin in Nonmuscle Invasive Bladder Cancer
	New Protocol Summary: The purpose of this study is to explore an alternative treatment called recombinant Bacillus Calmette-Guerin (rBCG) for patients with high-risk non-muscle invasive bladder cancer, especially given the worldwide shortage of the standard Tice BCG treatment. rBCG is a genetically modified strain of BCG. Two gene modifications have been implemented in BCG to improve its immunogenicity and safety: First, a gene of <i>Listeria monocytogenes</i> (<i>L. monocytogenes</i>) coding for the protein listeriolysin (LLO) was integrated into the genome of BCG. LLO is responsible for the formation of pores in the phagosome after endocytic uptake of the bacterium, enabling the antigens of the bacterium to move into the cytosol. Second, BCG's gene for urease C was inactivated. Urease C catalyzes the hydrolysis of urea into carbon dioxide and ammonia, thus creating a basic environment. Its inactivation was necessary as the aforementioned listeriolysin is optimally active under acidic conditions. Training: Complete Applicable Section of the NIH Guidelines: Section III-C Containment Conditions: BSL2 Special Provisions: Hospital/Clinic Infection Control precautions Discussion: • A Committee member asked as if any AEs were associated with bacteremia? The reviewer was unsure. • A Committee Member asked if the IP was pan-sensitive? The Reviewer answered it was similar to TICE and could be treated effectively. Voting: A motion was made to approve the protocol and was seconded. Total 10, For 10, Opposed 0, Abstain 0

b. Basic Protocols

1. Howitt, M.	[5529] The impact of tuft cells and type 2 immunity on parasite and bacterial associated changes to the intestinal epithelium and mucosal immune system.
	Revision: Updated Personnel, Updated Description, Updated Agents Used, Updated Funding, Updated Attachments Summary: The lab would like to understand how type 2 immunity and tuft cells regulate M cells (microfold cells) and antigen transcytosis across the intestinal epithelium. M cells in the follicle-associated epithelium overlying gut-associated lymphoid tissue mediate the uptake and transcytosis of luminal antigens and microbes. Salmonella enterica serovar Typhimurium and rotavirus both commonly cross the epithelium through M cells, so the lab will use them as tractable models to measure how type 2 immune signaling alters M-cell number and function and the M-cell-dependent transcytosis of these microbes. Training: Complete Applicable Section of the NIH Guidelines: III-D Containment Conditions: BSL2 Special Provisions: enhanced decontamination New Agent Added: Rotavirus mouse strain rD6/2-2g, Salmonella Typhimurium SL1344 Facility Visit: August 3, 2026 Discussion: • A Committee Member asked about sharps precautions; the Reviewer responded the lab is using engineered sharps with a disposal container located next to the work area. Voting: A motion was made to conditionally approve the protocol and was seconded. Condition for Approval: Provide the biogram for the Salmonella strain. Total 10, For 10, Opposed 0, Abstain 0
2. Hie, B.	[6007] Production and Use of Pseudotyped Lentiviral, Vesicular Stomatitis Virus (VSV), and Moloney Murine Leukemia (MMLV) particles
	Revision: Updated Personnel, Description, Agents Used, rDNA, animals/cells, and risk Summary:

	The lab is designing de novo antibodies against filovirus glycoproteins displayed on lentivirus and moloney murine leukemia particles and replication-deficient Vesicular Stomatitis Virus (VSV) lacking endogenous glycoprotein and incorporating a fluorescent reporter gene, in an effort to develop broadly neutralizing therapeutics. Filovirus envelope glycoprotein plasmid: Filovirus genes (Marburg GP (Angola/Musoke strains) or Ebola GP (Zaire/Reston/Sudan strains)). Training: Complete Applicable Section of the NIH Guidelines: III-D Containment Conditions: BSL2+ Special Provisions: enhanced decontamination New Agent Added: Vesicular Stomatitis Virus (G deleted protein), Moloney Murine Leukemia Virus(viral vector) Facility Visit: July 27, 2026 Discussion: • A Committee member asked if the AI is only used to design the antibody;The Reviewer responded this was correct. The panel discussed adding language to the approval letter to clarify that any other use of AI would need to be submitted via revision. • A Committee member asked if the VSV was produced as a viral vector system or a replication-competent virus? The Reviewer confirmed this was only a viral vector. This led to a Committee discussion on how the virus or viral vector was generated. The Committee requested further information before proceeding with review and voting. The Protocol was Tabled.
3. Kim, P.	[6131] Viral Entry and Neutralization Assays Revision: Updated Description, Updated Agents Summary: The lab would like to develop a protein subunit measles vaccine to protect the most vulnerable groups during measles outbreaks. Babies under the age of 1 year and immunocompromised individuals are ineligible to receive live attenuated vaccines, like Measles, Mumps, and Rubella Virus Vaccine Live (MMR II), therefore these outbreaks have put these groups severely at risk. They will be using the fusion protein from the measles virus surface. They will stabilize the protein, delete the furin cleavage site, display on ferritin nanoparticles, and test durability to generate a safe and effective vaccine. The lab will compare their protein subunit measles vaccines with the currently

	available MMR II. Training: Complete Applicable Section of the NIH Guidelines: III-D Containment Conditions: BSL2 Special Provisions: N/A New Agent Added: MMR II Vaccine Facility Visit: July 29, 2026 Discussion: • A Committee Member asked about the experience of the researcher; the Reviewer responded they had experience with other agents, in vivo work and will complete specific ABSL2 training. • A Committee Member asked if AI was used in design, and if so, how many recombinant versions were produced. The Reviewer noted they would follow up and this was made a condition of approval. Voting: A motion was made to conditionally approve the protocol and was seconded. Condition for Approval: Clarify if and how AI was used, and how many recombinant protein versions were made. Total 10, For 10, Opposed 0, Abstain 0
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The meeting was adjourned at 6:08 PM.